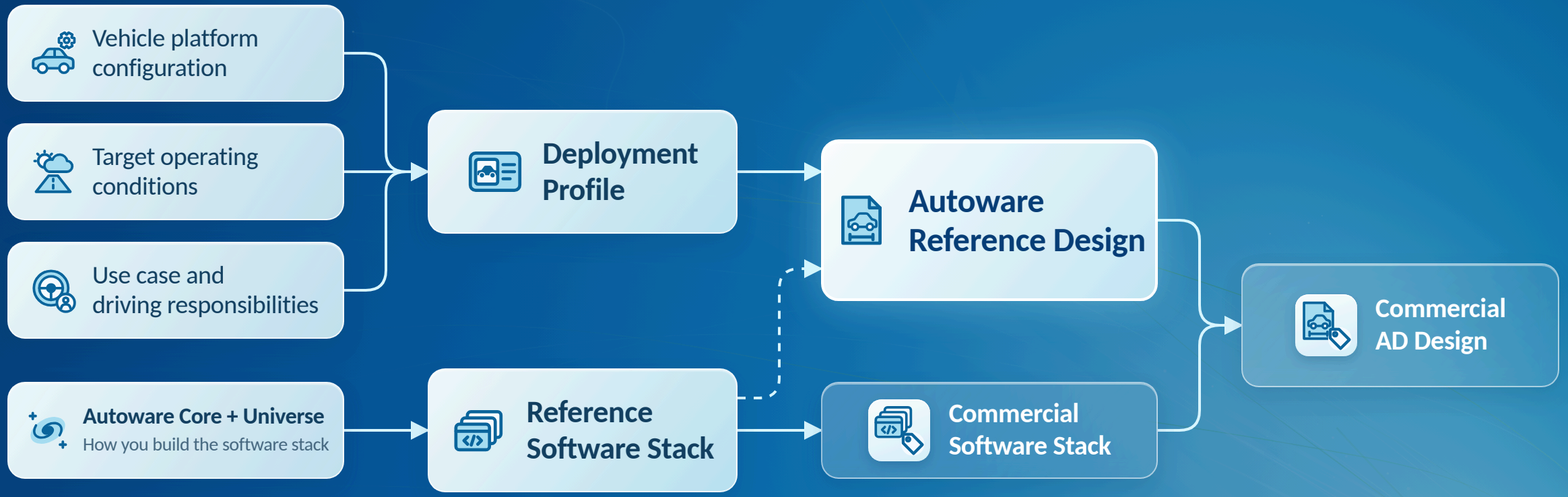
Autoware reference design

A complete system blueprint with reusable parts.



Deployment Profile + Reference Software Stack = **Reference Design**

The deployment profile

The platform and operating target for a system.



A **Deployment Profile** describes the intended application and operating conditions for a specified vehicle and hardware configuration.

Software candidates can share a profile, provided their requirements fit.

🚗 Vehicle platform

Vehicle, sensors, compute, actuation, and interfaces.

🌍 Operating conditions

Roads, speed, weather, lighting, and other ODD boundaries.

🎯 Use case

Driving functions and driver or operator responsibilities.

Campus shuttle Illustrative profile	LiDAR, cameras, GNSS/IMU Specified vehicle and compute	Paved campus routes Daylight and dry weather	≤ 15 km/h

The profile states the target. The integrated design defines its ODD and records what evaluation supports.

The reference software stack

A software configuration Autoware publishes and maintains as a recommended baseline.

Rule-based Planning	Localization and perception, rule-based planning, control, and supporting modules.
Diffusion Planning	A specified diffusion planner, required inputs, trajectory checks, and control.
Meteor-based	A specified Meteor model, runtime, adapters, and supporting Autoware modules.
Vision Pilot	AutoSteer, AutoSpeed, AutoDrive, and explicitly selected supporting modules.
Fusion Pilot	AutoE2E, input processing, runtime, output adapters, and supporting modules.

> **A name resolves to a configuration.**
>
> Modules, model versions, parameters, interfaces, and platform requirements.

These are candidate baselines. Their complete manifests and supported deployment profiles remain to be specified.

Reference design families

Organize the catalog around applications. Keep implementation choices explicit.



Low-speed shuttle

Campus and local passenger transport



Highway assistance

Driver assistance on highway routes



Urban robotaxi

Passenger transport within a defined ODD

Family

Low-speed shuttle

Reference design

Campus shuttle

Deployment profile + reference software stack

Documented variants

- Rule-based planning
- Diffusion planning
- Meteor-based

A family groups related designs. Each design can have a baseline and documented variants.

A catalog with independent attributes

Application, ODD, vehicle, sensors, software, and evaluation each have their own field.

Family	Target ODD	Platform	Sensors	Software	Evaluation
Low-speed shuttle	Paved campus, daylight, dry, ≤15 km/h	Shuttle	LiDAR, cameras, GNSS/IMU	**Rule-based Planning**	Proposed
Low-speed shuttle	Paved campus, daylight, dry, ≤15 km/h	Shuttle	LiDAR, cameras, GNSS/IMU	**Diffusion Planning**	Proposed
Low-speed shuttle	Campus target; model requirements to confirm	Shuttle	Sensor requirements to confirm	**Meteor-based**	Proposed
Highway assistance	Highway target; ODD to specify	Passenger vehicle	Camera-based; inputs to specify	**Vision Pilot**	Proposed
Urban robotaxi	Urban target; ODD to specify	Passenger vehicle	Camera and LiDAR; details to specify	**Fusion Pilot**	Proposed

Illustrative catalog entries. Proposed operating targets, compatibility, and evaluation remain to be established.

A campus shuttle reference design

One application can support several explicitly defined system configurations.

Variant	Deployment profile	Software choice
A	Campus shuttle platform and operating target	**Modular software with rule-based planning**
B	Campus shuttle platform and operating target	**Modular software with a diffusion planner**
C	Campus shuttle platform and operating target, subject to Meteor's platform requirements	**Meteor-based configuration**
D	Campus shuttle with cameras and radar, targeting campus operation	**Camera–radar perception with rule-based planning**

> **Evidence belongs to a defined configuration.**
>
> Each variant records its version, integration requirements, and evaluation scope.

Illustrative combinations. Changes to hardware, software, or operating conditions require an assessment of the affected evidence.

The complete system blueprint

A documented, versioned reference design makes the system reproducible and assessable.

Purpose	Driving functions, use case, automation target, and human responsibilities.
Operating domain	Conditions, boundaries, assumptions, and exclusions.
Platform	Vehicle, sensors, compute, actuation, and interfaces.
Software	Selected modules, models, versions, parameters, and dependencies.
Integration	Calibration, timing, maps, resources, and deployment instructions.
Evaluation	Criteria, tested configurations, results, and known limitations.

A reference design can begin as a draft. Its status must distinguish intended capability from demonstrated results.

Decisions for the board

Agree on the artifacts and their relationships, then define concrete examples.

1. **System boundary**
   Reference Design combines a Deployment Profile and a Reference Software Stack.
2. **Reusable definitions**
   Share platform, ODD, and software definitions across designs and families.
3. **Ownership and publication**
   Specify the minimum contents and maintainers for each official reference.
4. **Initial examples**
   Describe rule-based, learned-planner, and camera-based configurations with one template.

A **family** organizes. A **profile** sets the target.
A **stack** supplies the software. A **design** brings them together.